Ex.no:6	# DATA VISUALIZATION USING POWER BI (VEHICLE ANALYSIS)

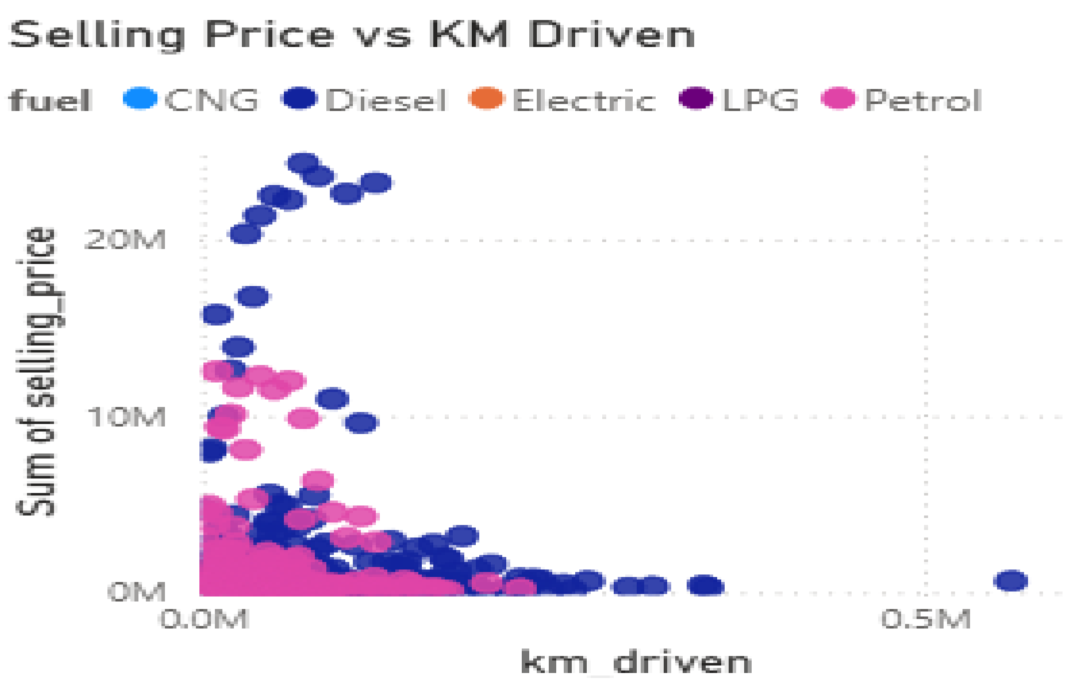
Aim:

To clean, preprocess, and transform the raw vehicle analysis dataset, load the clean data model into Power BI, and build an interactive visual dashboard to analyze vehicle prices, usage, fuel types, transmission types, seller types, and ownership.

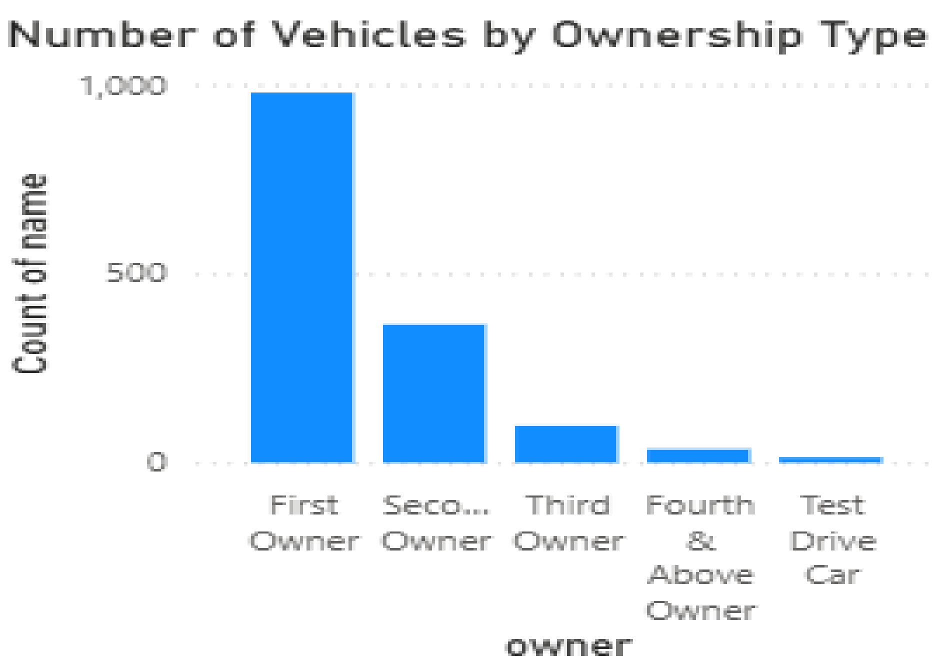
Algorithm:

1. **Import Dataset**: Load the **vehicle analysis CSV dataset** into Power BI Desktop using **Get Data → Text/CSV**.
2. **Remove Duplicate Rows:** Identify and remove duplicate records from the dataset to ensure each vehicle entry is unique.
3. **Handle Missing Values:** Identify columns containing null/missing values and replace them appropriately using 0, median, or mode, depending on the column.
4. **Handle Data Type Errors:**
   Replace invalid or non-numeric values such as "NA" or blank values in numerical columns with appropriate values such as **0**.
5. **Format Data Types:**
   Set numerical fields such as **Price, Mileage, Engine, Power, and Year** to the appropriate numeric data types.
   Set categorical fields such as **Car Name, Fuel Type, Seller Type, Transmission, and Owner Type** as **Text**.
6. **Clean and Transform Data:**
   Standardize categorical values and remove unnecessary columns that are not required for vehicle analysis.
7. **Load Clean Data:**
   Apply the transformations using **Power Query → Close & Apply** to load the cleaned vehicle dataset into Power BI.
8. **Create Dashboard:**
   Create an interactive **Vehicle Analysis Dashboard** by combining the required metrics and visualizations.

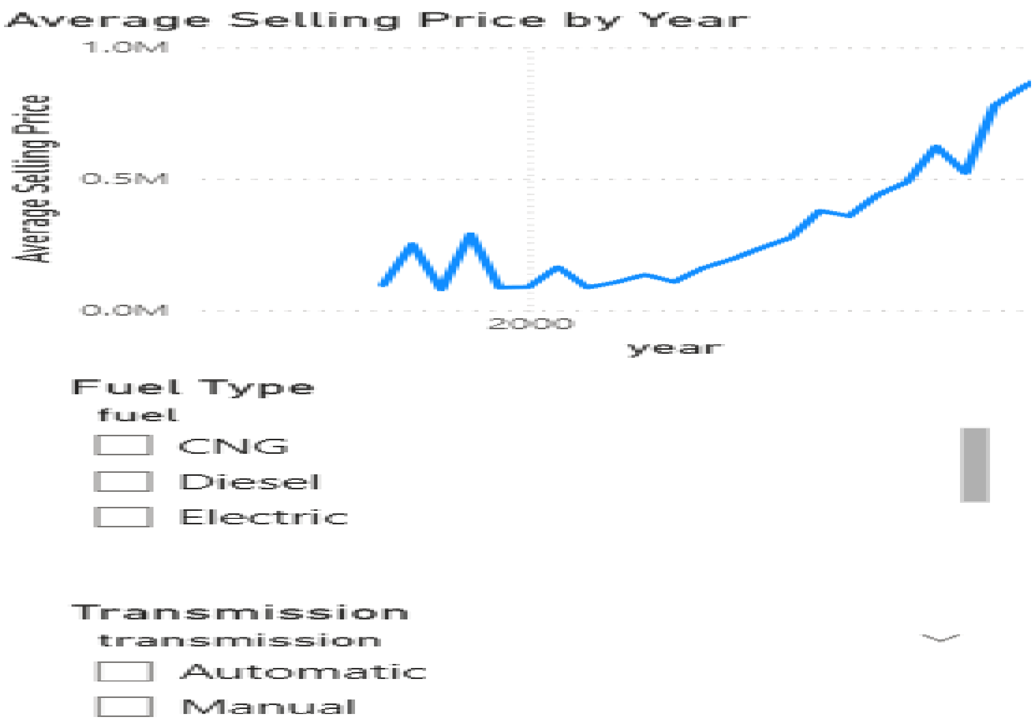
Output:



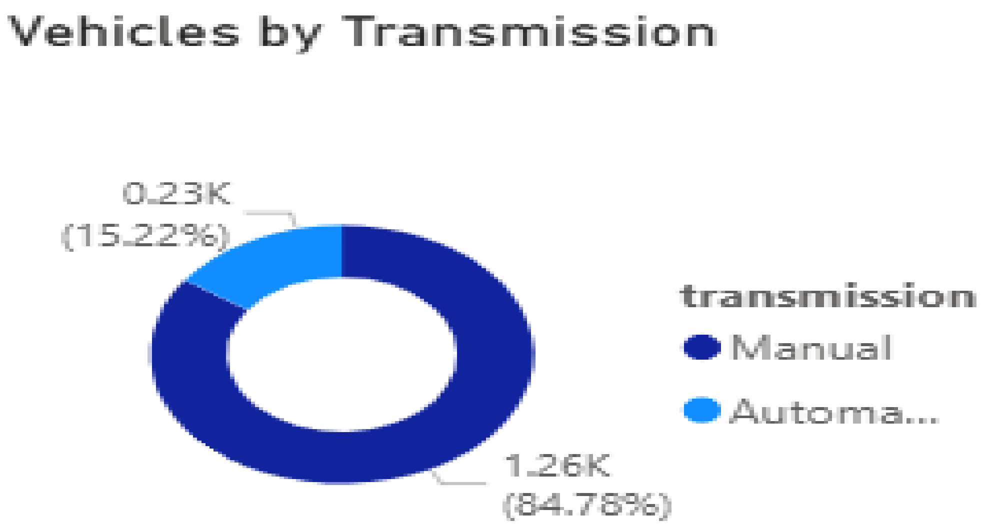
Selling Price vs KM Driven
fuel CNG Diesel Electric LPG Petrol
20M
10M
0M
Sum of selling_price
0.0M
0.5M
km_driven



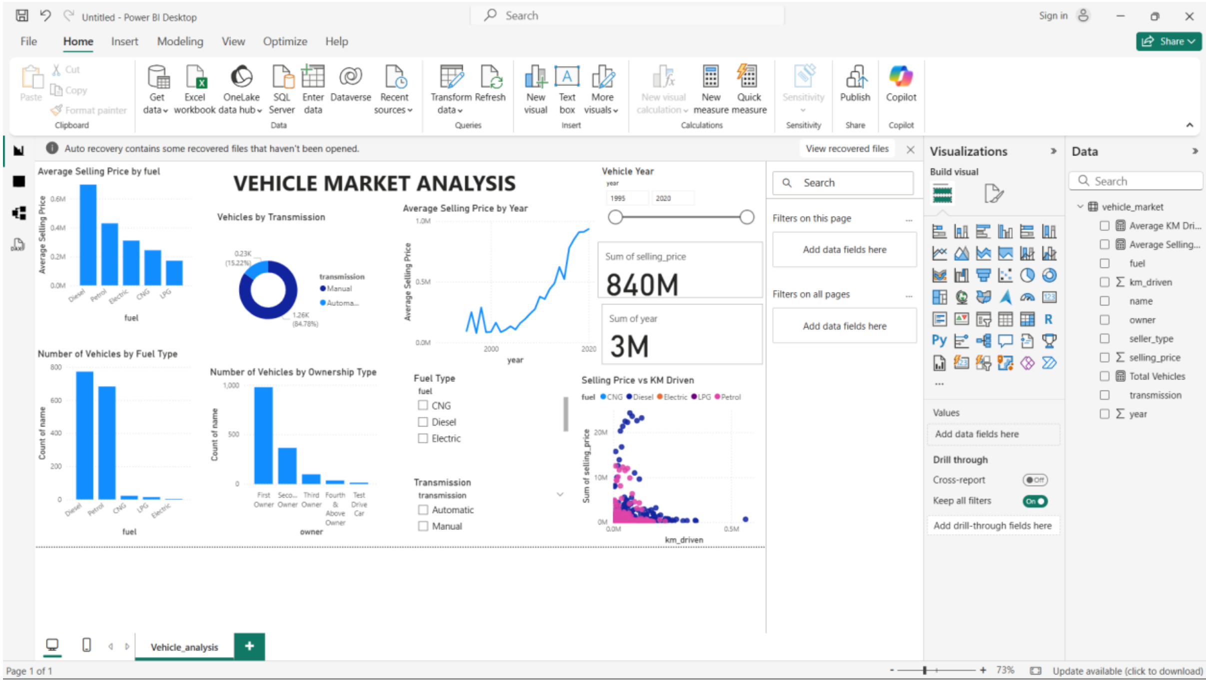
Number of Vehicles by Ownership Type
1,000
500
0
Count of name
First Owner
Seco... Owner
Third Owner
Fourth & Above Owner
Test Drive Car
owner



Average Selling Price by Year
1.0M
0.5M
0.0M
Average Selling Price
2000
year
Fuel Type
fuel
CNG
Diesel
Electric
Transmission
transmission
Automatic
Manual



Vehicles by Transmission
0.23K
(15.22%)
1.26K
(84.78%)
transmission
Manual
Automa...



Untitled - Power BI Desktop
VEHICLE MARKET ANALYSIS
Average Selling Price by fuel
Vehicles by Transmission
Average Selling Price by Year
Vehicle Year
Sum of selling_price
840M
Sum of year
3M
Number of Vehicles by Fuel Type
Number of Vehicles by Ownership Type
Fuel Type
Transmission
Selling Price vs KM Driven
Vehicle_analysis

Result: